CLIP-Based Multi-Modal Feature Learning for Cloth-Changing Person Re-Identification

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***Abstract*—Contrastive Language-Image Pre-training (CLIP) has achieved remarkable results in the field of person re-identification (ReID) due to its excellent cross-modal understanding ability and high scalability. Since the text encoder of CLIP mainly focuses on easy-to-describe attributes such as clothing, and clothing is the main interference factor that reduces the recognition accuracy in cloth-changing person ReID (CC ReID). Consequently, directly applying CLIP to cloth-changing scenario may be difficult to adapt to such dynamic feature changes, thereby affecting the precision of identification. To solve this challenge, we propose a CLIP-based multi-modal feature learning framework (CMFF) for CC ReID. Specifically, we first design a pose-aware identity enhancement module (PIE) to enhance the model's perception of identity-intrinsic information. In this branch, to weaken the interference of clothing information, we apply a ranking loss to minimize the difference between appearance and pose in the feature space. Secondly, we propose a global-local hybrid attention module (GLHA), which fuses head and global features through a cross-attention mechanism, enhancing the global recognition ability of key head information. Finally, considering that existing CLIP-based methods often ignore the potential importance of shallow features, we propose a graph-based multi-layer interactive enhancement module (GMIE), which groups and integrates multi-layer features of the image encoder, aiming to enhance the contextual awareness of multi-scale features. Extensive experiments on multiple popular pedestrian datasets validate the outstanding performance of our proposed CMFF.**

***Index Terms*—CC ReID, contrastive language-image pre-training, graph attention network.**

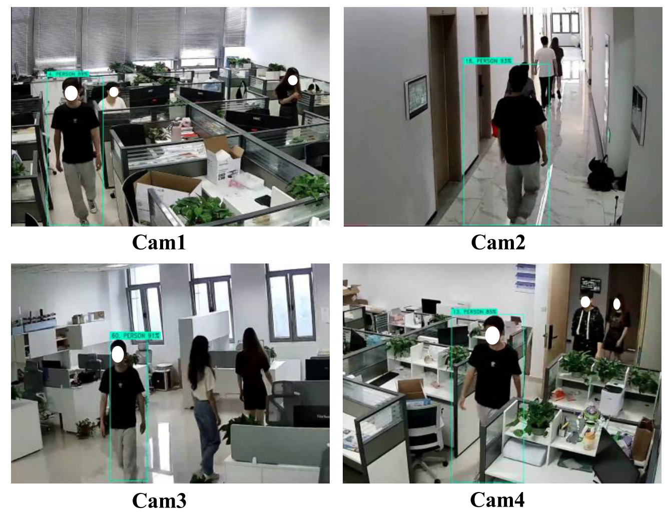

Fig. 1. Retrieval examples for short-term (e.g., in a same day) person ReID.

I. Introduction

PERSON re-identification (ReID) aims to retrieve target pedestrians across non-overlapping camera networks (as illustrated in Fig. 1). During recent years, it has become a crucial technique in various public safety and monitoring, including criminal suspect tracking, missing person localization, and intelligent traffic management. However, in real-world scenarios, recognition accuracy is affected by many factors, such as illumination changes, object occlusion, cross resolution, and modality differences. Therefore, researchers have proposed a large number of methods to deal with the above challenges [1], [2], [3], [4]. For example, Zhang et al. [1] proposed a progressive difference elimination model that reduces the impact of cross-modal and intra-class differences between visible light and infrared modalities on recognition by fusing visible and infrared stripes and generating high-quality transitional modality images. Li et al. [2] achieved resolution-invariant image representation through generative adversarial networks, successfully recovering the lost details in low-resolution input images. Lin et al. [3] proposed a temporal camera contrast learning framework, aiming to reduce the intra-ID discrepancy caused by background shift in unsupervised person re-identification and accurately extract invariant semantic cues in pedestrian images.

While these methods demonstrate promising results under controlled conditions, they fundamentally rely on the assumption of stable appearance features – an assumption frequently violated in practical long-term surveillance scenarios. As shown in Fig. 2, clothing changes over days or months significantly alter pedestrian appearance, posing substantial challenges for conventional appearance-based ReID systems.

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Fig. 2. An example of correct matches of a pedestrian in clothing-changing scenarios.

This limitation underscores the urgent need to develop clothing-invariant ReID methods that maintain robust performance during long-term real-world deployment.

To deal with the interference of clothing changes on identity information, researchers have conducted in-depth studies in the field of clothing-change ReID (CC ReID). Existing CC ReID methods can be primarily classified into two groups: one focuses on mining clothing-independent biomodal information, such as body contour, gait and posture [5], [6], [7]. To mitigate the influence of clothing information, the other methods separate clothing from the non-clothing human parts [8], [9], [10], [11] and focus on non-clothing parts to improve recognition accuracy. However, methods that rely on auxiliary biological modalities will greatly limit identity features in the case of poor image quality (such as severe occlusion or low-light environment). In addition, merely removing the clothing part will not only damage the integrity of the non-clothing area, but also lead to the loss of crucial identity information. Recently, contrastive language-image pre-training [12] (CLIP) has attracted widespread attention due to its excellent zero-shot learning capabilities and has achieved superior progress in multiple computer vision fields such as object classification [13], anomaly detection [14], and semantic segmentation [15]. In the person ReID tasks, CLIP improves the model's capability to capture pedestrian identity features by analyzing the intricate semantic connection between images and textual descriptions, thus boosting both the precision and textcolorbluerobustness of recognition. For instance, Yan et al. [16] proposed a CLIP-driven framework CFine, which aims to exploit the powerful multi-modal knowledge of CLIP to mine fine-grained and discriminative details for effective global alignment. In addition, to deal with the partial loss of appearance details due to occlusion, Cui et al. [17] proposed a prompt-guided feature disentanglement method (ProFD) that generates robust local features through text prompt activation and utilization of the rich knowledge pre-trained in the model.

Although the CLIP model demonstrates great potential in multi-modal representation learning, it also encounters significant challenges that cannot be ignored when applying it

Fig. 3. Example of traditional image-text person ReID, where yellow box represents correct matches and red boxes denote incorrect ones.

directly to the CC ReID task; we use CLIP-ReID [18] as our backbone network and evaluating on the clothing dataset Celebreid [19]. The visual outcomes of some experiments are presented in Fig. 3, and it is evident that although CLIP is able to generate text cues that are closely related to pedestrian clothing, its cross-modal capability does strengthen the connection between visual content and text descriptions. However, the model shows obvious deficiencies in handling identity information interference caused by clothing changes. This limitation mainly stems from CLIP's pre-training mechanism, which tends to emphasize salient features of visual elements such as clothing, while ignoring subtle features, such as facial features or body posture that are critical to identity recognition.

To tackle the aforementioned challenges, we put forward a CLIP-based multi-modal feature fusion network (CMFF) for CC ReID, which combines image, text, and biological (head, posture) modalities to enhance the model's robustness. Firstly, we propose a pose-aware identity enhancement module that leverages the pose estimation model HRNet [20] to capture the detailed pose information of pedestrians. By combining the pose information with the global visual features obtained from CLIP, the model's ability to perceive the inherent characteristics of pedestrian identities can be enhanced. Among them, to minimize the impact of clothing changes on recognition accuracy, we use the ranking loss function to fine-tune the model to effectively narrow the semantic distance between posture and appearance features. Furthermore, given that the head region contains richer identity information than other body parts, we adopt an innovative global-local hybrid attention to promote the interaction between global features and head features. This fusion strategy not only allows to guide and strengthen the emphasis on head information from a global perspective, but also enables head features to be contextually enhanced based on global information, aiming to significantly improving feature representation capability. Finally, to enhance the information flow between features at each layer of the image encoder, we employ a graph attention structure to carefully group and integrate features from multiple layers, thereby optimizing the semantic richness and discriminability of the overall feature hierarchy.

Our primary contributions are outlined in the following points.

- We propose a CLIP-based multi-modal feature fusion network (CMFF) for CC ReID that combines image, text, and biological modalities information to strengthen the model's robustness in identity recognition.

- We propose a posture-aware identity enhancement module (PIE), designed to enhance the model's capability of perceiving inherent identity information of individuals by fusing global appearance features with structural pose features.
- We propose a global-local hybrid attention module (GLHA) to contextually enhance head features by leveraging global information, aiming to guide and strengthen the attention on head information from a global perspective.
- We propose a graph-based multi-layer interactive enhancement module (GMIE) that applies a graph attention mechanism to finely group and fuse multi-layer image encoder features, effectively enhancing the semantic richness and discriminability at the feature level, significantly boosting the model's overall performance.

II. Related Works

A. Cloth-Changing Person Re-Identification

Person ReID aims to retrieve images of the same pedestrian from different camera perspectives and plays a vital role in public safety applications. However, most existing methods [21], [22] depend on clothing information as the primary identifier, ignoring the fact that pedestrians clothing changes in real scenes. Consequently, cloth-changing person ReID has attracted a large number of scholars to study as an emerging topic, with its primary objective being to identify and learn stable identity features that are independent of clothing.

1) Auxiliary Biological Information Based Methods: Thanks to the collection of multiple cloth-changing pedestrian datasets [19], [23] [24], [25], CC ReID research has made great progress. To reduce the visual interference caused by clothing changes, mainstream methods use auxiliary information such as gait [5], [26], silhouette/shape [6], [27], face [28], [29] and posture [7] to learn clothing independent identity features. Feng et al. [27] proposed AGS-Net, an attention-guided two-stream framework that utilizes visual and contour information from RGB images and silhouette sketches as discriminative features. This two-branch structure focuses on identity-related features while minimizing sensitivity to clothing changes using an attention module. Nguyen et al. [7] proposed a contrastive clothing and pose generation framework (CCPG) to jointly learn clothing-independent features through appearance and shape, and generate cross-identity clothing and pose transformation images to enrich training samples, thereby supervising the learning of discriminative features. Wu et al. [28] recovered the lost facial details from the teacher network and propagate them to a smaller network, aiming to mine sensitive identity cues in faces. Compared with short-term ReID methods, these methods have significantly improvements in clothing changing scenes, but they rely too much on the acquisition of auxiliary information, which limits the scalability and generalization of the model and also leads to a significant increase in computational costs. In addition, the high requirements for image quality of this method mean that once pedestrians are blocked or in a dark light environment, the generated auxiliary information will inevitably contain noise, affecting the stability of training.

2) Clothing Decoupling Based Methods: Another mainstream methods is to decouple the clothing parts of the human body from the non-clothing parts, enabling the model to concentrate on learning clothing-independent features [8], [9], [10], [11]. Eom et al. [8] proposed an identity shuffled adversarial network that using only identity labels without any auxiliary supervision signals to separate identity-related and irrelevant features from portraits. Zheng et al. [10] proposed a joint learning framework that separates the appearance and shape features of each image and exchanges them within and between identities to achieve data augmentation. SAVS [11] located the human body and clothing regions through human semantic segmentation, masked clues related to clothing appearance, and only focuses on visual semantic knowledge that is insensitive to view/pose changes. However, due to the lack of exact ground truth guidance, the entire feature decoupling process is performed implicitly, which makes the model inevitably disturbed by clothing features when extracting clothing-independent features. In addition, the high computational overhead incurred by these methods when processing large-scale data remains a problem that needs to be optimized.

B. CLIP in Person Re-Identification Task

Contrastive Language-Image Pre-training [12] has demonstrated remarkable zero-shot learning capabilities by adopting a contrastive learning strategy on a large-scale multi-modal dataset. Its core idea is to map images and descriptive text into the same vector space through contrastive learning, optimize the similarity between images and text and learn the representation of the association between visual content and language descriptions. As a result, it is not limited to fixed datasets and predefined categories when processing visual tasks, but can also understand concepts or objects that have not been seen during training. This training strategy gives CLIP strong generalization capabilities, enabling it to effectively perform zero-shot inference on a range of visual tasks, including fine-grained object classification [13], anomaly detection [14], and semantic segmentation [15].

Recently, researchers have explored the use of CLIP in the domain of person ReID [18] and have made significant progress in scenarios such as visible-infrared [30], occlusion [31] and lifelong learning [31]. Acknowledging that high-level semantics of pedestrian appearance are consistent across modalities, Yu et al. [30] proposed a CLIP-driven semantic discovery network, which seeks to bridge the gap between visible and infrared modalities by combining visual features with high-level semantics. To address the challenge of cross-domain knowledge mismatch in lifelong person ReID under different clothing states, LReID-Hybrid [32] leveraged the consistency and generalization of the text space and introduced a framework that efficiently aligns, transfers, and accumulates knowledge, creating an image-text-image closed loop. He et al. [31] designed a multi-granularity contrastive consistency alignment framework, aiming to semantically align occluded visual effects and query text by leveraging intra-granular/inter-granular visual text representations.

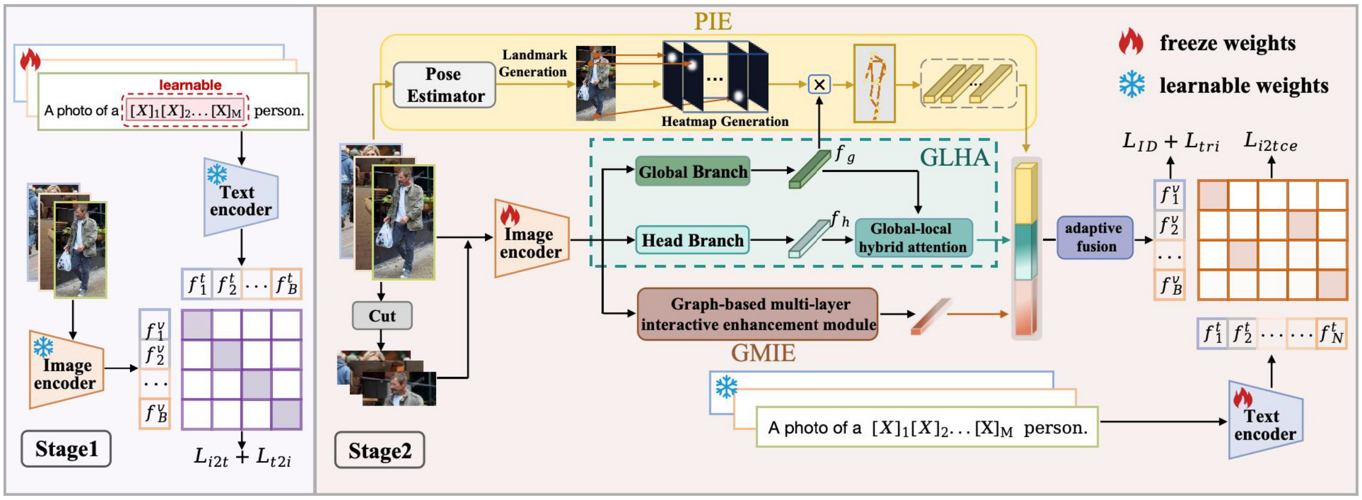

Fig. 4. The CLIP-based multi-modal feature learning framework (CMFF) we proposed consists of two stages.The first training phase fixes the text encoder and image encoder and optimizes a set of learnable text tags to generate text features. In the second stage, the visual encoder adopts a three-branch structure to enhance the accuracy of identity recognition. Specifically, posture-aware identity enhancement module (PIE) improves the perception of identity features by introducing posture information; global-local hybrid attention module (GLHA) targets crucial head information to boost global recognition; graph-based multi-layer interactive enhancement (GMIE) module groups and integrates the image encoder's multi-layer features to enhance the contextual awareness of multi-scale features.

Later, the CLIP model has also been explored for clothing-changing scenarios, but this attempt has encountered some challenges that cannot be ignored: since the CLIP model tend to emphasize visible and describable attributes during training, which may lead to the model to over-focusing on clothing features in CC ReID scenarios, while ignoring intrinsic features that are more closely related to personal identity, such as the head and pedestrian profile.

To this end, we propose a CLIP-based multi-modal feature learning framework (CMFF), which adopts a multi-branch structure to mine biological modalities such as posture and head to enhance the recognition of identity information. Furthermore, we use the graph structure to fuse the multi-layer features of the image encoder, enhancing the contextual understanding ability of multi-scale features.

III. Method

A. Overview

In our framework, we adopt CLIP-ReID [18] as the backbone network, which enhances the missing text information via a collection of pre-trained learnable text tags. The entire method is split into two stages, as illustrated in Fig. 4.

In the initial training stage, we adopt learnable text tags associated with each identity to handle ambiguous text descriptions and ensure that these descriptions are unique for each ID. Specifically, the text description $T(\cdot)$ is formatted as a photo of a $[X]_1[X]_2[X]_3 \ldots [X]_M$ person", where each $[X]_m$ ($m \in \{1, \ldots, M\}$) is a learnable text tag, and M represents the total number of text tags. In this stage, we refine model by image-to-text contrastive loss L_{i2t} and text-to-image contrastive loss L_{t2i}, aiming to enhance the consistency of cross-modal representation:

$$L_{i2t} = -\log\left(\frac{\exp(s(V_i, T_i))}{\sum_{a=1}^{B}\exp(s(V_i, T_a))}\right), \tag{1}$$

and

$$L_{t2i} = -\log\left(\frac{\exp(s(V_i, T_i))}{\sum_{a=1}^{B}\exp(s(V_a, T_i))}\right), \tag{2}$$

where $s(\cdot, \cdot)$ denotes the cosine similarity, which is used to calculate the similarity between image embedding V_i and text embedding T_i in the cross-modal space. In addition, it is also used to compare the similarity of V_i with other text embeddings T_a in the same batch.

In the second stage, we first fuse the pose information with global visual features extracted by CLIP to improve the model's capability to perceive the inherent characteristics of pedestrian identity. Then, we propose a global-local hybrid attention module that emphasizes head information based on a global perspective and enhances head features in context through global information. Finally, we apply a graph attention mechanism to finely group and integrate multi-layer image encoder features to enhance the model's contextual awareness and feature-level semantic richness.

B. Posture-Aware Identity Enhancement Module

Since CLIP's text encoder primarily focuses on easily describable attributes (e.g., clothing), it struggles to capture subtle but crucial identity-specific features such as body structure and posture. To address this limitation and reduce the model's dependency on clothing-related features, we propose a pose-aware identity enhancement module (PIE) that aims to leverage the inherent body structure information of pedestrians to enhance the model's perception of persistent identity markers that remain stable across clothing variations.

Pedestrian posture plays a key role in embodying key structural information about the human body, such as characteristics that remains invariant across changes in attire, lighting conditions, and viewing perspectives. By combining it with global visual features, we aim to mitigate the influence of

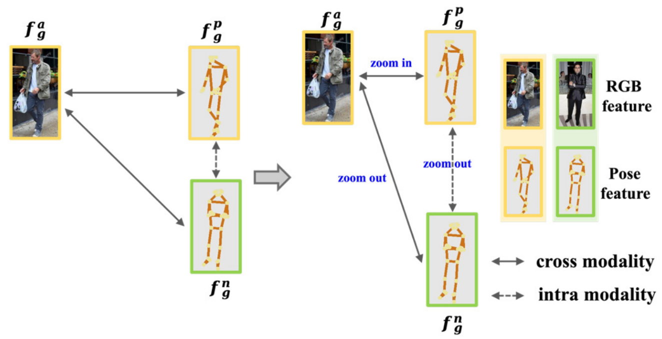

Fig. 5. Visualization of the ranking loss, where the same color borders represent the same ID. The purpose of the loss is to make RGB-pose pairs of the same identity closer and separate RGB-pose pairs of different identities.

clothing variations on global visual representations, thereby enhancing the model's stability and robustness under significant changes in pedestrian appearance. Specifically, for a given pedestrian image x_i, we employ the pose estimation model HRNet [20] to locate the coordinates of K body key points. Based on these coordinates, we generate individual heatmaps $H_k(k = 1, 2, \ldots, K)$ as follows:

$$H_k(x, y) = \exp\left(-\frac{(x - x_k)^2 + (y - y_k)^2}{2\sigma^2}\right), \tag{3}$$

where (x, y) represents the spatial coordinates of a point in the heatmap, σ represents standard deviation of the Gaussian kernel, controlling the spread of the distribution, and (x_k, y_k) is the location of the k-th key point in the image. Furthermore, we aggregate all K individual heatmaps into a combined posture heatmap H_m:

$$H_m = \sum_{k=1}^{K} H_k, \tag{4}$$

where $H_m \in \mathbb{R}^{W \times H}$, with W and H denoting the width and height of the heatmap, respectively. To integrate local pose information with global visual features, we first expand the global image features $f_g \in \mathbb{R}^D$ to match the spatial dimensions of the posture heatmap H_m, and then perform element-wise multiplication to obtain the pose-enhanced features f_p:

$$f_p = H_m \odot Expand(f_g, W, H), \tag{5}$$

where $\odot$ denotes element-wise multiplication, $Expand(\cdot, W, H)$ is a function that reshapes the dimension of a feature to $W * H$. The fusion process enriches the feature details and improves contextual relevance by combining pose and global visual information.

Inspired by the contrastive loss based on silhouette sketches and RGB images in [6], we introduce a ranking loss L_{rank}, which aims to reduce the dependence of appearance features on clothing information by minimizing the distance between appearance features and pose features, thereby enhancing the model's attention to the core information of pedestrian identity. The specific calculation process is shown in Fig. 5, and in a batch, there are N images and their corresponding N posture heat maps. The distance between the RGB features and posture features (positive sample pairs) belonging to the same ID should be smaller than the distance between the global features and posture features (negative sample pairs) belonging to different IDs. Meanwhile, the distance between posture features of different IDs needs to be increased, which ensures that the model can focus more on the inherent characteristics of pedestrians rather than the changeable appearance information during the recognition process. Therefore, we propose the inter-class ranking loss L_{cross} and the intra-class ranking loss L_{intra}:

$$L_{cross} = \frac{1}{N} \sum_{i=1}^{N} \max\left(0, \delta_1 + \|f_{g,i}^a - f_{p,j}^+\|_2 - \|f_{g,i}^a - f_{p,t}^-\|_2\right), \tag{6}$$

$$L_{\text{intra}} = \frac{1}{N} \sum_{i=1}^{N} \max\left(0, \delta_2 + \|f_{p,i}^a - f_{p,j}^+\|_2 - \|f_{p,i}^a - f_{p,t}^-\|_2\right), \tag{7}$$

where $\|\cdot\|_2$ represents the Euclidean distance, δ_1 and δ_2 are preset interval thresholds, which are used to guarantee that the distance between the positive sample feature and the anchor point is significantly less than the distance between the negative sample and the anchor point. Afterwards, we optimize the model performance by integrating L_{cross} and L_{intra} losses into the total ranking loss function and using weight coefficients λ_1 and λ_2 to adjust their weights in the total loss. The specific formula is as follows:

$$L_{\text{rank}} = \lambda_1 L_{cross} + \lambda_2 L_{\text{intra}}. \tag{8}$$

C. Global-Local Hybrid Attention Module

Although CLIP is good at capturing the global features of an image, it often ignores the importance of local regions, especially the head region, which usually contains more key and rich identity recognition features than other parts of the body. To overcome this limitation and enhance the model's ability to focus on key identity regions, we propose a global-local hybrid attention module to enhance the model's ability to focus on head features by effectively fusing with global information. For a given image x_i, we first apply the pose estimation model [20] to accurately locate the key points of the human head and shoulders. Then, based on these key points, we use the advanced image segmentation algorithm Mask R-CNN to accurately extract the head and neck area from the entire image. Next, we feed both full pedestrian image and its corresponding head image into image encoder separately. Considering the difference in dimensions between the two inputs, we perform adaptive pooling operations on the global and head images in the global and head branches respectively, thereby obtaining the global feature f_g and head feature f_h.

The global-local hybrid attention is shown in Fig. 6. We first perform different linear transformations on f_g and f_h to map them into the form of query (Q), key (K) and value (V), respectively. Different from traditional self-attention mechanism, we use the query vector of f_g to pay attention to the key and value (V) vectors of f_h, and vice versa. Then, we combine these two weighted features through the concatenation operation

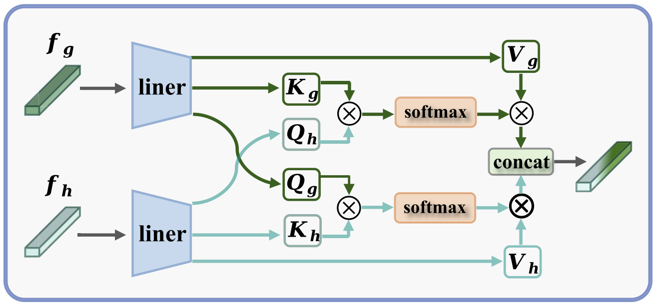

Fig. 6. Illustration of the global-local hybrid attention module (GLHA), which aims to enhance the importance of head information from global perspective.

$concat(\cdot,\cdot)$ to form the final fused feature F_{g-h}. The detailed expression is provided below:

$$A_{\text{g-h}} = \text{softmax}\left(\frac{Q_{\text{g}}K_{\text{h}}^T}{\sqrt{d}}\right) \odot V_{\text{h}}, \tag{9}$$

$$A_{\text{h-g}} = \text{softmax}\left(\frac{Q_{\text{h}}K_{\text{g}}^T}{\sqrt{d}}\right) \odot V_{\text{g}}, \tag{10}$$

$$F_{g-h} = concat(A_{\text{g-h}}f_g, A_{\text{h-g}}f_h). \tag{11}$$

In this way, our model can effectively capture the interdependence between head features and the global context of the whole body image.

D. Graph-Based Multi-Layer Interactive Enhancement Module

Most existing CLIP-based methods use the final layer features of the image encoder (ViT) as the main visual representation, which may ignore the valuable information contained in the intermediate layers. To address this under-utilization of multi-scale features and enhance the contextual understanding ability of the model, we propose a graph-based multi-layer interactive enhancement module (GMIE).

The detailed architecture of GMIE is shown in Fig. 7. Firstly, we extract the output features of each layer from the ViT model and denote $X^{(l)} \in \mathbb{R}^{N\times D}$ as the output features of the l th layer, where $l \in \{1, 2, \ldots, 12\}$, N and D represents the sequence length and feature dimension. Prior to inputting these features into the graph attention module, we perform layer normalization on the output features of each layer:

$$\hat{X}^{(l)} = \frac{X^{(l)} - \mu^{(l)}}{\epsilon + \sigma^{(l)}}, \tag{12}$$

where ϵ is a small constant that prevents the denominator from being zero, $\mu^{(l)}$ which $\sigma^{(l)}$ represent the mean and standard deviation of $X^{(l)}$ respectively, and are computed as follow:

$$\mu^{(l)} = \frac{1}{D}\sum_{d=1}^{D} X_{:,d}^{(l)}, \tag{13}$$

$$\sigma^{(l)} = \sqrt{\frac{1}{D}\sum_{d=1}^{D}(X_{:,d}^{(l)} - \mu^{(l)})^2}. \tag{14}$$

Subsequently, we map the normalized features to graph nodes and divide the nodes into three groups ($t = 3$) according to the low to high levels of Transformer. We empirically set the number of groups to t=3 as it provides an optimal balance between feature diversity and information completeness, which is further validated through ablation studies in Section IV. We then establish the relationship between nodes based on the features. The edge weight e_t^{ij} is determined by the similarity between nodes, reflecting the relative importance of node i in relation to node j in t-th group, and the calculation formula of e_t^{ij} is as follows:

$$e_t^{ij} = \frac{\exp(\text{sim}(\hat{X}_t^{(i)}, \hat{X}_t^{(j)}))}{\sum_{k\in N_i}\exp(\text{sim}(\hat{X}_t^{(i)}, \hat{X}_t^{(q)}))}, \tag{15}$$

where $sim(\cdot,\cdot)$ denotes cosine similarity between two nodes, and N_i represents the set of neighbors of node i. Then, we calculate the attention coefficient α_t^{ij} between the two nodes, which quantifies the influence of the neighboring node $\hat{X}_t^{(i)}$ on node $\hat{X}_t^{(j)}$ during the feature update process:

$$\alpha_t^{ij} = \frac{e_t^{ij}\exp(\text{LeakyReLU}(\mathbf{a}^T[W_t^i \| W_t^j]))}{\sum_{q\in\mathcal{N}_i} e_t^{ij}\exp(\text{LeakyReLU}(\mathbf{a}^T[W_t^j \| W_t^q]))}, \tag{16}$$

where W is a learnable linear transformation matrix, a is a learnable vector for calculating attention and $\|$ represents the concatenation of vectors. We then update the node features h_t^i with the adjusted attention coefficients:

$$h_t^i = ReLU\left(\sum_{q\in\mathcal{N}_i}\alpha_t^{iq}\hat{X}_t^{(i)}\right), \tag{17}$$

where $ReLU(\cdot)$ represents nonlinear activation function. Afterwards, we aggregate each group of node features through maximum pooling to obtain the final feature h_t of each group:

$$h_t = maxpooling(h_t^1, h_t^2, h_t^3, h_t^4), t \in \{1, 2, 3\}. \tag{18}$$

Finally, we concatenate the outputs of each group processed by the graph attention network to construct the final feature representation F_{graph} of GMIE, aiming to leverage features of each layer in the image encoder to improve the model's robustness in complex scenarios.

E. Adaptive Fusion and Model Optimization

1) Adaptive Feature Fusion: To maximize the utilization of information from different branches and optimize the final feature representation, we perform a adaptive fusion operations on the output features of the three branches. Specifically, we initialize the weights of each branch through fully connected layer and normalize them with softmax function. According to these generated weights, we perform weighted aggregation of the three feature vectors, and adaptively adjusted the weights during back propagation:

$$F_{fused} = w_{pose}f_p + w_{g-h}F_{g-h} + w_{graph}F_{graph}. \tag{19}$$

2) Model Optimization: In the first stage, we employ the image-to-text contrast loss L_{i2t} and text-to-image contrast loss L_{t2i} to reduce cross-modal discrepancy between images and textual descriptions. The overall loss function is formulated as follows:

$$L_{stage1} = L_{t2i} + L_{i2t}. \tag{20}$$

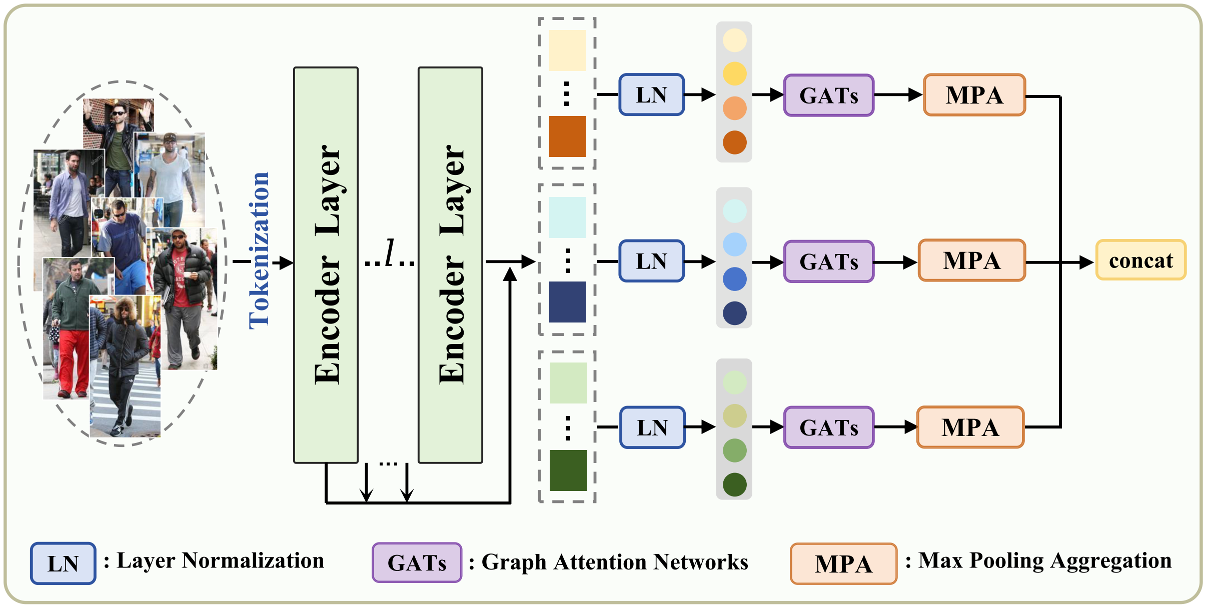

Fig. 7. Illustration of graph-based multi-layer interactive enhancement module.

In the second stage, we first apply a ranking loss L_{rank} in the pose-aware identity enhancement moduleto minimize the discrepancy between appearance and pose features. Furthermore, we adopt identity loss L_{ID} and triplet loss L_{tri} to achieve further optimization of the model:

$$L_{\text{ID}} = -\sum_{i=1}^{C} y_i \log(p_i), \tag{21}$$

$$L_{\text{tri}} = \frac{1}{N}\sum_{i=1}^{N} \max\left(0, m + d_{ap} - d_{an}\right), \tag{22}$$

where C denotes number of categories, y_i is the one-hot encoding of the true label, p_i is the predicted probability of the corresponding category of the mode. N refers to the number of samples in same batch, d_{ap} represents distance between the anchor sample and the positive sample, and d_{an} indicates the distance from the anchor sample to the negative sample.

To fully exploit the potential of CLIP, we adopt an image-to-text cross-entropy loss to ensure that image features closely correspond to the corresponding text descriptions while keeping distance from text features of other irrelevant identities:

$$L_{\text{i2tce}(i)} = -\sum_{k=1}^{N} \log\left(\frac{e^{s(V_i,T_{yk})}}{\sum_{y_a=1}^{N} e^{s(V_i,T_{ya})}}\right). \tag{23}$$

where V represents the overall visual features fused in the second stage, and $S(\cdot,\cdot)$ is used to calculate the similarity score.

Thus, the overall loss function for the second stage is expressed as follows:

$$L_{stage2} = L_{rank} + L_{ID} + L_{tri} + L_{i2tce}. \tag{24}$$

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: To assess the effectiveness of our proposed CMFF method, we perform experiments on four publicly available cloth-changing pedestrian datasets: VC-Clothes [23], Celeb-reID [24], Celeb-reID-light [19], and NKUP [25]. In particular, for the VC-clothes dataset, we set two scenarios: one is "general", which includes both clothing changes and clothing consistency scenes; the other is "cloth-changing", which only includes clothing changes. The characteristics of each dataset are summarized in Table I.

2) Evaluation Metrics: Following previous methods, we adopt the first-rank hit rate (Rank-1) and mean average precision (mAP) as metrics to assess the efficacy of our proposed CMFF.

B. Implementation Details

In this study, we use ViT-B-16 in the CLIP ReID framework as the backbone network. The images are first resized to 256 × 128 resolution to accommodate the network input requirements. To improve the generalization of the model, we apply a series of data augmentation techniques, including random horizontal flipping, random erasing, and edge padding. During data loading, we use a hybrid sampler that combines softmax and triplet loss, processes 4 instances per batch and uses 8 worker threads for parallel data processing. The training process of this study is optimized in two stages.

In the first stage, we use Adam optimizer with a batch size of 128, a base learning rate of 3.5 $\times 10^{-4}$, and a training cycle of 60 epochs. In the second stage, the training cycle is extended to 120 epochs, while the batch size remains unchanged. The base learning rate is reduced to 0.5 $\times 10^{-5}$, the number of iterations is set to 10, and both weight decay and bias weight decay are configured at 0.1 $\times 10^{-3}$.

TABLE I

INTRODUCTION TO THE DATASET USED IN THE EXPERIMENT, WHERE 'SC' REPRESENTS THE SAME CLOTHING AND 'CC' REPRESENTS DIFFERENT CLOTHING

Dataset	Source	Train(ID/Image)	Test(ID/Image)		Cameras	Data Style
			Query	Gallery		
VC-Clothes	Synthetic	256/9,449	256/1020	256/8,591	4	SC/CC
NKUP	Real	40/5,336	39/332	67/4,070	15	CC
Celeb-reID	Internet	1,052/34,186	420/2,972	420/11,006	-	CC
Celeb-reID-light	Internet	100/887	100/934	590/10,842	-	CC

TABLE II

COMPARISON WITH SOTA METHODS ON VC-CLOTHES (%)

Methods	Assistance	General		Cloth-Changing	
		Rank-1	mAP	Rank-1	mAP
Short-Term Based Methods					
PCB (ECCV 18) [33]	None	87.7	74.6	62.0	62.2
DG-Net (CVPR 19) [10]	None	94.1	93.8	76.8	68.4
ISP (ECCV 20) [34]	Parsing	94.5	94.7	72.0	72.1
Cloth-Changing Based Methods					
FSAM (CVPR 21) [35]	Parsing+Pose	94.7	94.8	78.6	78.9
CAL (CVPR 22) [36]	Clothes ID	92.9	87.2	81.4	81.7
GI-ReID (CVPR 22) [37]	Parsing+Gait	-	-	64.5	58.7
ACID (TIP 23) [38]	None	95.1	94.7	84.3	74.2
IMS-GEP (TMM 23) [39]	None	-	-	81.8	81.7
MSD-Net (ICMR 24) [40]	Parsing	94.1	93.9	84.1	83.4
GLH (Neurocom. 24) [41]	None	94.2	90.2	86.4	86.1
DSIFLF (ACM MM 24) [42]	Parsing	**96.8**	93.6	94.3	90.2
baseline	description	95.7	93.4	92.3	88.5
CMFF (ours)	description + Pose	96.3	**94.5**	**94.8**	**90.4**

C. Performance Evaluations and Comparison

In this experiment, we compare our method with the current state-of-the-art (SOTA) methods on the VC-Clothes, NKUP, Celeb-reID, and Celeb-reID-light datasets. In addition, we further compare and analyze the performance differences of these methods with or without auxiliary information.

1) Results on VC-Clothes: Consistent with previous studies, our comparative experiments on the VC-Clothes dataset adopt the following two settings: 1) **General setting**, where gallery sets include both samples with clothing changes and samples with the same clothing; 2) **Cloth-Changing setting**, where the gallery sets just contain samples that have undergone clothing changes. As shown in Table II, our method achieves the highest performance in both settings. Compared with the second-best method DSIFLF [42], our method improves Rank-1 and mAP by 0.5% and 0.2% in Cloth-Changing scenarios, and improves mAP by 1.3% in the General scenarios. Among the contrasting methods, GLH [41] utilizes a multi-branch architecture to capture both global and local identity information, and trains local body parts by clustering the generated pseudo labels for cross-image comparison. However, this strategy that relies on automatically generated pseudo labels may introduce errors due to inconsistent label quality, thus affecting the accuracy and robustness of the model. FSAM [35] enhances clothing-independent features by transferring fine-grained body shape knowledge between two stream branches. DSIFLF [42] uses a human body parsing model to erase clothing parts, suppresses the fluctuations caused by clothing texture in the identity feature space, and reduces the dependence on interference factors during discriminative learning by generating adversarial interference factor decoupling networks. GI-ReID [37] incorporates gait recognition as an auxiliary task and utilizes unique gait information to guide model in learning features that are independent of clothing. Although these methods improve the accuracy of feature extraction through single biomodal information, they also significantly rely on the quality and consistency of the data, which limits their generalization ability in new environments or changing conditions. In contrast, our method combines pose estimation and global features, fully leveraging the identity recognition information of the head, which substantially enhances the model's capacity to capture inherent identity information.

2) Results on Celeb-reID and Celeb-reID-Light: We perform comparative experiments with state-of-the-art methods on the Celeb-reID and Celeb-reID-light datasets, with detailed results provided in Table III. Obviously, ours surpasses all comparative methods on both datasets. MADE [50] extracts personal attribute descriptions through attribute detection model and masks out clothing and color information. However, when clothing attributes are closely combined with other body features, masking clothing attributes alone is not enough to completely eliminate all clothing-related information. AFD-Net [44] and SAFR [11] use generative adversarial networks to separate identity-related and irrelevant features. Among them, the decoupling and reconstruction of AFD-Net rely

TABLE III
COMPARISON WITH SOTA METHODS ON CELEB-REID AND CELEB-REID-LIGHT DATASETS (%)

Methods	Assistance	Celeb-reID		Celeb-reID-light	
		Rank-1	mAP	Rank-1	mAP
Short-Term Based Methods					
MGN (ACMMM 18) [43]	None	49.0	10.0	13.9	21.5
PCB (ECCV 18) [33]	None	37.1	8.2	9.0	16.7
DG-Net (CVPR 19) [10]	None	50.1	10.6	23.5	12.6
Cloth-Changing Based Methods					
AFD-Net (IJCAL 21) [44]	GAN	52.1	10.6	22.2	11.3
MBUNet (TIP 22) [45]	Pose	55.3	12.1	33.9	21.3
MCSC (TIP 24) [46]	None	57.8	13.0	29.2	16.2
SAFR (TIP 22) [47]	GAN	56.0	14.2	29.5	16.7
FIRe2 (TIP 24) [48]	None	64.0	18.2	-	-
SAVS (TNNLS 23) [11]	Parsing	65.9	21.3	29.5	16.7
3DInvarReID (ICCV 23) [49]	3D	65.5	18.4	42.2	25.7
GLH (Neurocom. 24) [41]	None	<u>69.0</u>	23.6	-	-
MADE (arxiv 24) [50]	description	-	-	<u>69.6</u>	<u>52.1</u>
baseline	description	67.8	<u>28.1</u>	67.4	46.9
ours	description+ Pose	**69.7**	**30.2**	**71.3**	**52.3**

TABLE IV
COMPARISON WITH SOTA METHODS ON NKUP DATASET (%)

Methods	Assistance	Rank-1	mAP
Short-Term Based Methods			
PCB (ECCV 18) [33]	None	18.7	14.1
MGN (ACMMM 18) [43]	None	20.6	16.1
Cloth-Changing Based Methods			
FIRe2 (TIFS24) [48]	None	22.4	16.0
MVSE (ACMMM 21) [51]	Parsing	23.8	17.9
SAVS (TNNLS 23) [11]	Parsing	25.3	18.6
UCAD (IJCAI 22) [52]	None	25.0	16.9
CCUP (arviv 24) [53]	None	26.4	17.8
IGCL (TPAMI 23) [54]	Parsing	**28.8**	**28.0**
baseline	description	25.3	18.6
ours	description+ Pose	<u>27.9</u>	<u>19.5</u>

on complex self-supervision, which increases the difficulty of training and is less efficient. Although SAFR learns disentangled features from random samples, the quality and diversity of input samples limit its generalization ability. 3DInvarReID [49] innovatively uses 3D body model to separate identity and non-identity features, but extreme posture changes and clothing diversity may reduce the accuracy and efficiency of its decoupling. In contrast, our approach employs a multi-branch architecture to extract critical identity cues from the head and posture, while leveraging a graph attention mechanism to aggregate multi-scale features, thereby significantly enhancing the model's performance in complex scenarios.

3) Results on NKUP: We conduct comparative experiments with existing methods on the NKUP dataset, and the results are shown in Table IV. In this experiment, our CMFF method obtains the second highest performance, with rank-1 and mAP 0.9% and 8.5% lower than IGCL [54] respectively. The main reason is that the images of the NKUP dataset are collected in a dark environment, and the image clarity is low. IGCL introduces clothing attention activation maps to mask clues related to the appearance of clothes, and enriches the expression of different postures under the same identity through image stitching technology, which can effectively extract human semantic information independent of the background and adapt to the posture changes of pedestrians. These strategies enable IGCL to show a strong advantage in processing low-definition images. CCUP [48] also demonstrated the effectiveness of synthetic data pre-training in CC-ReID tasks, where the pre-train-fine-tune framework leverages large-scale synthetic data to enhance the generalization ability of the model. However, the performance gap compared to IGCL suggests that synthetic pre-training may be of limited effectiveness for low-quality surveillance scenarios like NKUP, where field-specific adaptation methods (such as the attention mechanism and pose enhancement strategy of IGCL) show greater advantages when handling low-resolution images under challenging lighting conditions.

D. Ablation Study

1) Effectiveness of Graph Structure Grouping: We perform an in-depth analysis on the impact of feature grouping in GMIE module and set different grouping numbers $n = \{1, 2, 3, 4, 6\}$ for experiments. The results presented in Fig. 8 demonstrate that mode's performance fluctuates significantly with change of the number of groups. In particular, when n=1, the mAP of the model is always the lowest regardless of the nature of the dataset; while when $n = 3$, it performs optimally on all datasets. In addition, the number of groups that is too low or too high will cause performance degradation. The main reason is that when features are only divided into one group, the model that relies on a single fusion output may over-compress information, resulting in the loss of critical fine-grained features such as edges and textures, which are essential for identifying small differences in complex scenes. It is evident that a moderate increase in the number of groups (such as $n = 3$) can effectively retain and utilize features at different scales, balance the diversity and completeness of information through the complementarity of features within each group,

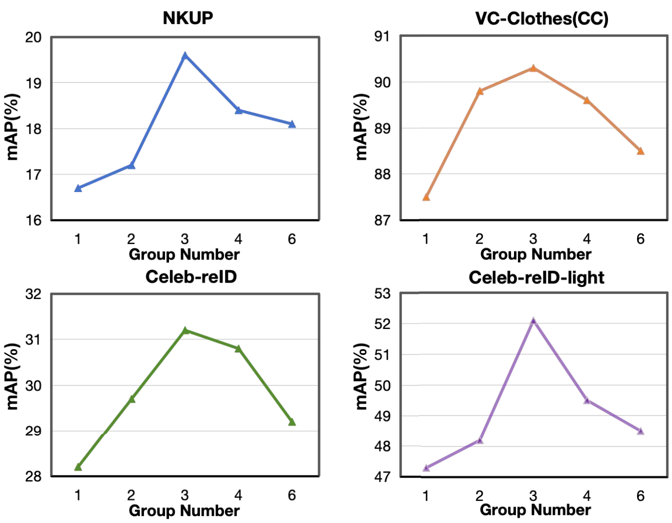

Fig. 8. Results with different numbers of groups on four cloth-changing pedestrian datasets.

TABLE V
COMPARISON WITH SOTA METHODS ON VC-CLOTHES, NKUP AND CELEB-REID DATASETS (%)

Methods	Datasets					
	VC-Clothes(CC)		NKUP		Celeb-reID	
	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
baseline	92.3	88.5	25.3	18.6	67.8	28.1
+PIE	92.5	88.9	25.7	19.1	67.5	28.7
+GLHA	92.8	89.3	26.4	19.2	68.1	29.2
+GMIE	93.1	89.9	26.1	18.9	68.3	29.5
+PIE+GLHA	93.5	89.6	27.3	19.2	69.5	29.7
+PIE+GMIE	92.7	89.2	26.8	18.7	68.2	29.5
+GLHA+GMIE	93.3	89.8	27.1	19.3	69.0	29.8
+PIE+GLHA+GMIE	**94.8**	**90.4**	**27.9**	**19.5**	**69.7**	**30.2**

and thus optimize the model performance. However, too many groups (such as $n = 4$ or 6) will lead to over-dispersion of features, resulting in insufficient information in each group, which is not conducive to effective learning and recognition, thereby reducing the overall performance. Therefore, choosing the appropriate number of groups is extremely critical to improving model accuracy and adaptability.

2) Effectiveness of Each Module: We assess the contribution of each branch in CMFF framework through experiments conducted on three cloth-changing datasets: VC-Clothes, NKUP and Celeb-reID. We set CLIP ReID with Vision Transformer as the backbone network as the baseline, and integrate Pose-aware identity enhancement module (PIE), global-local hybrid attention module (GLHA) and graph-based multi-layer interactive enhancement module (GMIE) on this baseline, as well as different combinations of these modules, to explore their specific impact on performance. Table V lists all the results, and its cumulative matching feature (CMC) curve is shown in Fig. 9. First, we can find that the introduction of a single module has achieved an improvement in accuracy relative to the baseline model on the three datasets. Specifically, the improvements of the PIE module on mAP are 0.4%, 0.5% and 0.2% respectively; the improvements of the GLHA module are 0.8%, 0.6% and 0.5%; and the improvements of the GMIE module are 1.4%, 0.3% and 1.4%. Secondly, PIE and GLHA, as biometric learning branches, focus on extracting pedestrian posture and head information respectively. The combination of the two achieves 1.3%, 2.0% and 2.7% Rank-1 improvement on the three datasets respectively compared with the baseline, proving complementarity of the two modules and their effectiveness in extracting discriminative features. On this basis, the GMIE module was introduced, and the Rank-1 accuracy was further improved by 1.3%, 0.6%, and 1.4%, verifying the effect of enhancing multi-scale context perception.

To highlight the individual contribution of each module more clearly, we further conduct ablation experiments by removing each component from the full model. The results demonstrate substantial performance degradation when excluding any single module: (1) removing the PIE module leads to Rank-1 accuracy drops of 1.5%, 0.6%, and 0.7% on the three datasets; (2) eliminating GLHA resulting in a more significant drop of 2.1%, 1.1%, and 1.5%; (3) removing GMIE causes a drop of 1.3%, 0.6%, and 1.2%. The findings reveal that GLHA contributes most significantly to overall performance, followed by GMIE and PIE, which is consistent with the intuition that head features maintain the highest reliability in clothing-changing scenarios. In addition, the consistent performance drops across all modules confirm their complementary nature and collective necessity for achieving optimal results.

We also visualize each branch and combination on Celeb-reID dataset, as illustrated in Fig. 10. It can be seen that the PIE, GLHA and GMIE modules alone outperform the baseline model, with the combination of PIE and GLHA significantly improving the matching rate for Ranks 1-10. Especially after the introduction of the GMIE module, the model accuracy is further improved, which is consistent with the results of our ablation experiments.

3) Effectiveness of λ_1 and λ_2: To comprehensively evaluate the robustness of the model, we conduct ablation experiments on the hyperparameters in the ranking loss function on the dataset NKUP, where both parameters varied between 0.1 and 1.0, as shown in Table VI. The experimental results show that the performance is optimal when $\lambda_1 = 0.5$ and $\lambda_2 = 0.5$, which indicates that balancing the weights of inter-class and intra-class ranking loss is crucial for effective feature learning. The model performs relatively stably in the range of $\lambda_1, \lambda_2 \in [0.3, 0.7]$, with performance fluctuations of less than 1%. However, when any parameter takes an extreme value (0.1 or 1.0), the performance will further degrade due to insufficient inter-class distinction or over-constrained intra-class variation. Therefore, the balanced parameter configuration ensures that the model can effectively coordinate inter-class separation and intra-class aggregation to achieve the optimal feature learning effect.

4) Effectiveness of Ranking Loss in PIE Module: To verify the effectiveness of our proposed ranking loss in the pose-aware identity enhancement (PIE) module, we conducted comparative experiments on two cloth-changing datasets: NKUP, and Celeb-reID. As demonstrated in Table VII, we systematically three variants: (1) using our proposed ranking loss

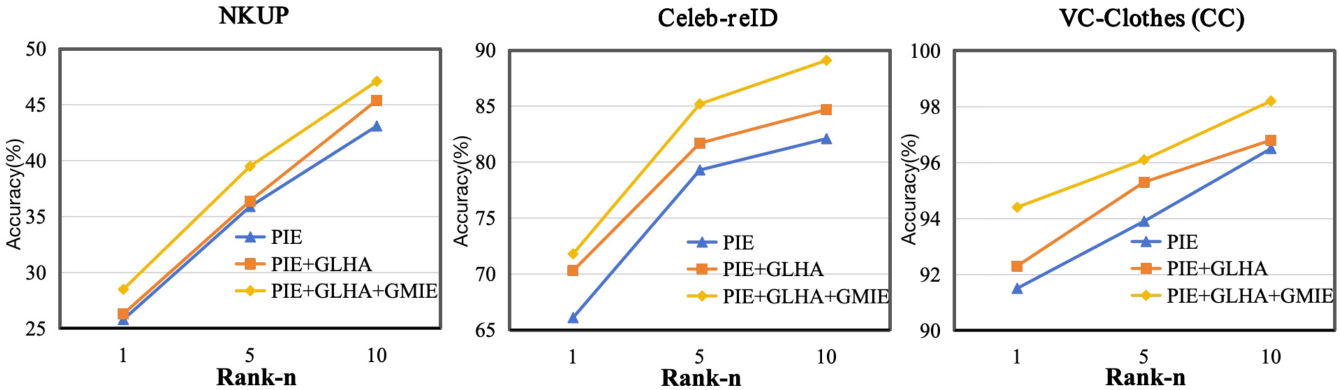

Fig. 9. Advantages of each module using CMC curves on NKUP, Celeb-reID and VC-Clothes datasets.

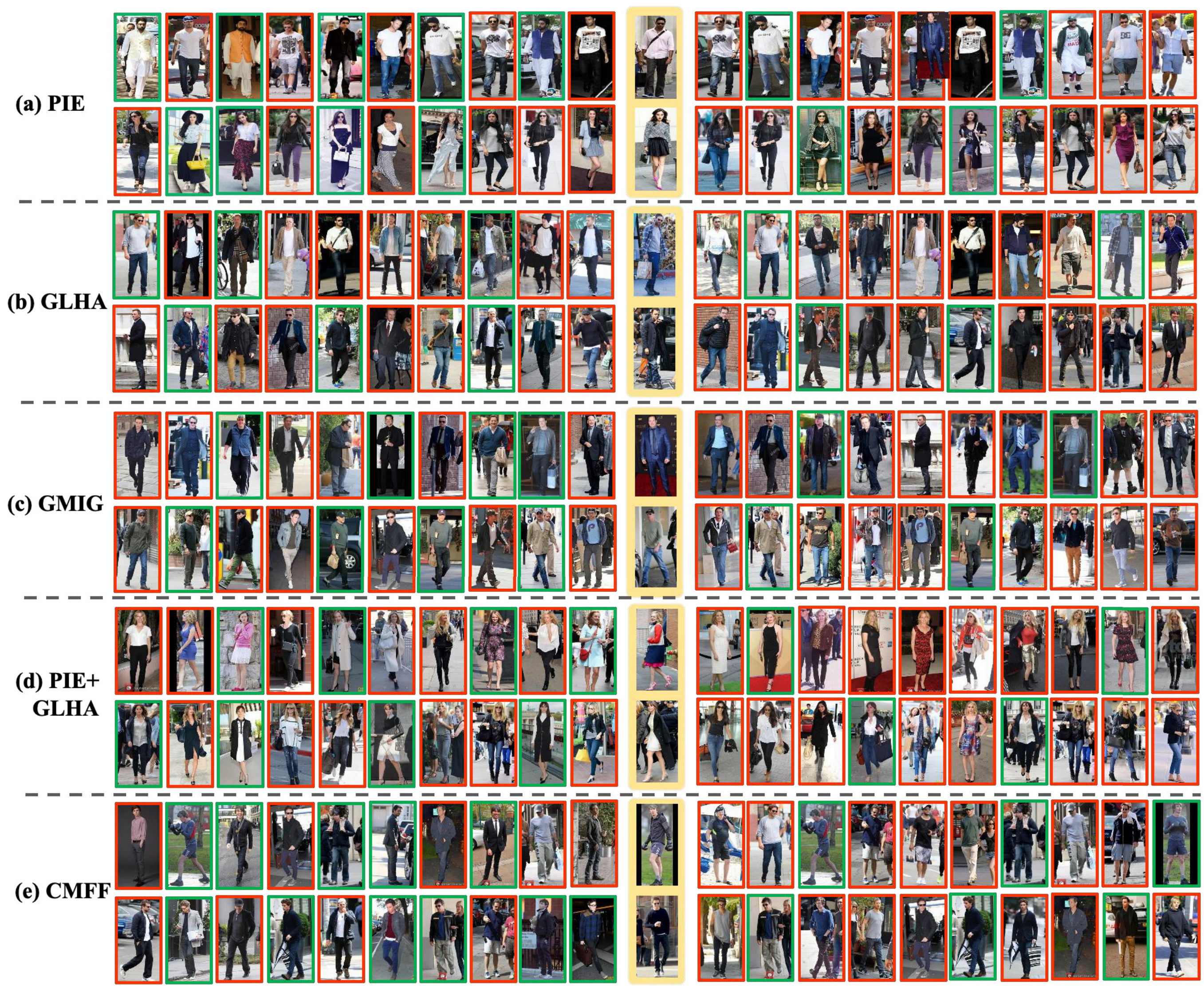

Fig. 10. Visualization of retrieval results on Celebreid dataset. The left side presents the results of our method (including various modules), and the right side shows the results of CLIP-ReID. Rank-1 to Rank-10 are reported. Positive matches are indicated by green rectangles, and negative matches are highlighted with red rectangles.

L_{rank}, (2) PIE using the standard L2 loss, and (3) the baseline. Experimental results show that our ranking loss consistently outperforms the other two alternatives on all datasets. It is worth noting that the L2 loss method only slightly improves over the baseline, and sometimes even performs worse, which indicates that simple distance minimization is not sufficient for

TABLE VI
PERFORMANCE OF DIFFERENT λ_1 AND λ_2 COMBINATIONS ON NKUP DATASET

$\lambda_1 \backslash \lambda_2$	0.1		0.3		0.5		0.7		1.0	
	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
0.1	26.8	18.9	27.1	19.0	27.3	19.2	27.0	18.8	26.7	18.6
0.3	27.2	19.1	27.5	19.3	27.7	19.4	27.4	19.2	27.1	18.9
0.5	27.6	19.3	27.8	19.4	**27.9**	**19.5**	27.7	19.3	27.4	19.1
0.7	27.4	19.2	27.6	19.3	27.8	19.4	27.5	19.2	27.2	18.9
1.0	27.0	18.8	27.3	19.1	27.5	19.2	27.3	19.0	26.9	18.7

TABLE VII
EFFECTIVENESS OF DIFFERENT LOSS FUNCTIONS IN PIE MODULE (%)

PIE Variant	NKUP		Celeb-reID	
	Rank-1	mAP	Rank-1	mAP
Baseline (CLIP-ReID)	25.3	18.6	67.8	28.1
PIE w/ L2 Loss	25.8	19.2	68.3	28.9
PIE w/ Ranking Loss (Ours)	**27.9**	**19.5**	**69.7**	**30.2**

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT FEATURE FUSION IN GMIE MODULE

GMIE Variant	NKUP		Celeb-reID	
	Rank-1	mAP	Rank-1	mAP
Simple Concatenation	27.4	19.1	68.7	28.9
Weighted Averaging	27.6	18.9	68.2	29.5
Ours	**27.9**	**19.5**	**69.7**	**30.2**

effective pose-appearance alignment. Therefore, our ranking loss design effectively minimizes the intra-class distance while maximizing the inter-class separation, which is crucial for cloth-changing person ReID.

5) Effectiveness of Graph Structure in GMIE Module: To verify the effectiveness of the graph structure in the multi-layer interaction enhancement (GMIE) module, we conducted comparative experiments on two clothing-changing datasets, NKUP and Celeb-reID, and the results are shown in Table VIII. We designed three variants for comparison: (1) GMIE using simple concatenation followed by linear projection, (2) GMIE using weighted average of multi-layer features, and (3) GMIE using our proposed graph-structure-based fusion method. Experimental results show that our method significantly outperforms other alternatives on both datasets. Although the simple concatenation method can integrate multi-layer information, it is constrained by feature redundancy and increased computational overhead, resulting in limited performance improvement. The weighted average method alleviates the dimensionality explosion problem to a certain extent, but it still cannot fully utilize cross-layer information due to the lack of ability to model complex spatial relationships between multi-scale features. In contrast, our graph structure effectively captures the spatial correlation of cross-layer features by adaptively learning the dependencies between features, thereby achieving more robust multi-scale context understanding in the cloth-changing person re-identification task.

6) Effectiveness of Bidirectional Attention in GLHA Module: To verify the superiority of the bidirectional attention

TABLE IX
PERFORMANCE COMPARISON OF DIFFERENT ATTENTION MECHANISMS IN GLHA MODULE

GLHA Variant	NKUP		Celeb-reID	
	Rank-1	mAP	Rank-1	mAP
fh→fg	26.1	18.0	68.3	29.4
fg→fh	27.6	18.9	68.5	29.5
Ours	**27.9**	**19.5**	**69.7**	**30.2**

mechanism we designed in the GLHA module, we conducted ablation experiments on the NKUP and Celeb-reID datasets, comparing three attention mechanisms: unidirectional $f_g \rightarrow f_h$ (global to head), unidirectional $f_h \rightarrow f_g$ (head to global), and our proposed bidirectional method. The experimental results in Table IX show that bidirectional attention can consistently achieve the best performance on both datasets, indicating that the mutual guidance between global features and head features is crucial for effective identity representation learning. Specifically, on the NKUP dataset, our bidirectional method outperforms the best unidirectional variant ($f_g \rightarrow f_h$) by 0.3% and 0.6% in Rank-1 and mAP, respectively. On the Celeb-reID dataset, bidirectional attention outperforms the unidirectional alternative by 0.2% and 0.7% in Rank-1 and mAP, respectively. This is sufficient to prove that the bidirectional design effectively combines the advantages of both directions, realizes comprehensive feature interaction and achieves superior recognition accuracy.

V. CONCLUSION

In this work, we have proposed a multi-modal feature learning method (CMFF) to tackle the challenges of cloth-changing person Re-ID. First, we designed a pose-aware identity enhancement module (PIE) that combines pose estimation with global feature analysis to strengthen the model's capacity of discriminating essential identity features and narrow the gap between appearance and pose-driven features through ranking loss. Next, we introduced a global-local hybrid attention module (GLHA) to highlight key head information in the global scope through head branch fusion with cross attention. Finally, we adopt a graph-based multi-layer interactive enhancement module (GMIE) to update and integrate multi-layer features of the image encoder to enhance the context-awareness of multi-scale features. Extensive experiments on several mainstream cloth-changing pedestrian datasets demonstrates the superior performance of our proposed CMFF.

Despite our approach achieving promising results, it still suffers from some limitations including computational complexity from the multi-branch architecture and dependency on pose estimation accuracy. For future work, we plan to integrate infrared modality to enhance performance in low-light conditions, develop lightweight variants for practical deployment, and explore video-based extensions with temporal consistency, which may provide additional discriminative cues for clothing-invariant ReID.

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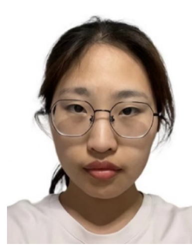

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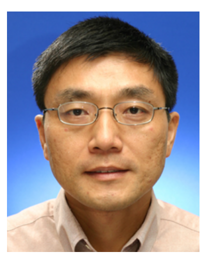

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